MILITARY SPECIFICATION SHEET

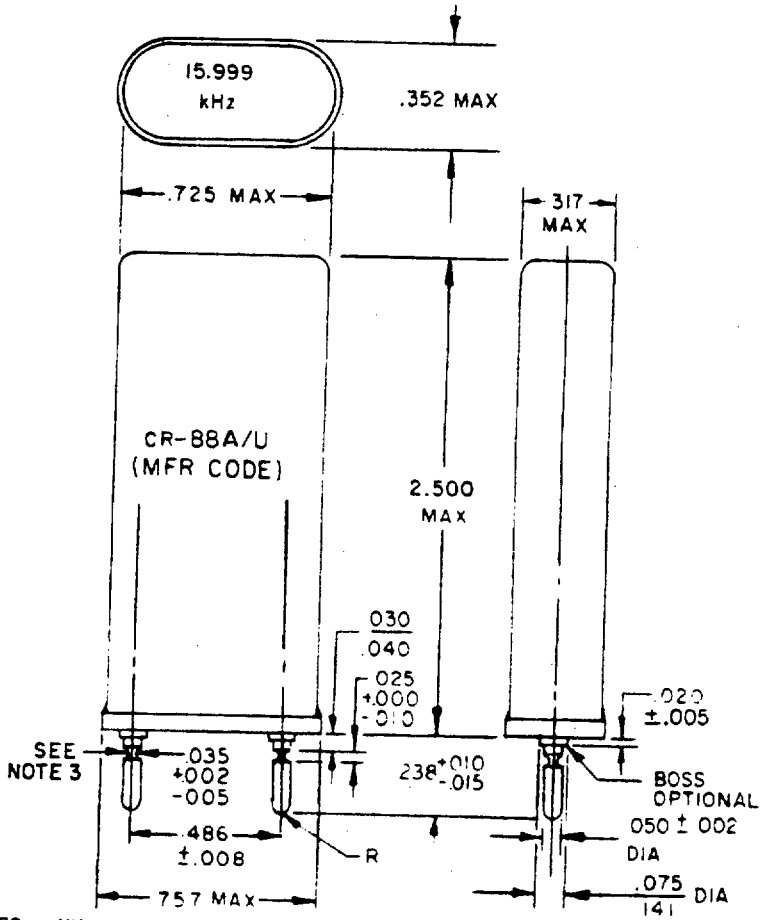
CRYSTAL UNIT, QUARTZ, CR-88A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

(F) **INACTIVE FOR NEW DESIGN**

Pertinent characteristics: 2.75 to 16 kHz; fundamental; noncontrolled; series.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.050	1.27	.352	8.94
.005	.13	.025	.64	.075	1.90	.486	12.34
.008	.20	.030	.76	.141	3.58	.725	18.42
.010	.25	.035	.89	.238	6.05	.757	19.23
.015	.38	.040	1.02	.317	8.05	2.500	63.50

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The pin undercut may be omitted.

FIGURE 1. Dimensions and configuration.

(F) denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	IAW figure and MIL-H-10056/4 (HC-13/U, modified. See note 3.)
Dimensions, marking, and configuration	See figure 1
Frequency range	2.75 to 16 kHz, inclusive
Frequency tolerances:	
Primary operating temperature range	±50 ppm
Secondary operating temperature range	±100 ppm
Equivalent resistance	100,000 ohms, maximum
Mode of oscillation	Fundamental
Operating temperature ranges:	
Primary	0°C to +65°C, inclusive
Secondary	-40°C to 0°C and 65°C to 70°C, inclusive
Government reference standard test set	TS-710/TSM
Rated drive level	0.1 mW
Calibration values	
Resistance	70,000 ohms
Resistor voltage drop	2.65 volts
Capacitance, shunt	kHz: 2.75 to 8 — pF: 15; 8+ to 16 — pF: 30 max
Shock: (specified pulse):	
Frequency change permitted	10 ppm
Equivalent resistance change permitted	±15 percent
Thermal shock:	
Frequency change permitted	10 ppm
Equivalent resistance change permitted	±15 percent
Vibration: 1/	
Frequency change permitted	10 ppm
Equivalent resistance change permitted	±15 percent
Bond strength	800 grams, minimum

1/ Method 201 of MIL-STD-202, amplitude 0.015 inch (0.030 inch total excursion).

Ⓕ QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTES:

1. The socket designed to mate with crystal holder HC-13/U is Part No. M12883/04-01 (see MIL-S-12883/4).
   For Navy use, other sockets listed in MIL-STD-242 may be used.

2. This crystal-unit type is to be used for special applications and for replacement purposes only.
   Specific approval of the Government is required prior to each intended use.
   See MIL-STD-683.

3. Crystal holder (see figure) is similar to type HC-13/U of MIL-H-10056/4. "Holder, Crystal, Type HC-13/U," except for cover height dimension.

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Custodian:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Air Force - 11
DLA - ES

User activities:
Navy - AS, MC, SH
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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